

Bachelor's Thesis Proposal

A Domain-Specific Language for Privacy-Aware Business Processes

Assessor: Dr. Hoang Nguyen

Co-assessor: To be decided

1 Motivation

Organizations describe how they operate in terms of *business processes*. The Business Process Model and Notation (BPMN) [2] is the modeling standard for business analysts and developers. It is supported by a large ecosystem of tools for drawing, validating, simulating, and even executing process models on workflow engines.

With the increasing interest in protecting users' personal data in software systems, a natural extension of BPMN is to integrate the declaration of privacy concepts into the processes. Ideally, the software designer should be able to define a business process model in which each activity or process is associated with a declared purpose. For example, a process for sending mass marketing emails to customers can be associated with a `MARKETING` purpose. This idea has been investigated and reported in the work of Basin et al. [1].

2 Objective

Since BPMN supports many features and has an extensive syntax, this thesis aims to provide a simpler abstraction for designing privacy-aware business process models. The objective of this thesis is to design a domain-specific language (DSL) for business process models and to implement a model transformation that automatically produces a standard BPMN model from it.

The DSL should allow designers to describe the basic concepts of a business process concisely while incorporating privacy concepts into the design.

3 Tasks

1. Study the approach proposed in the work of Basin et al. [1].
2. Study the BPMN specification and review existing textual notations for BPMN. Identify the subset of BPMN to be supported by the transformation.
3. Design the syntax of the DSL and the model transformation from the DSL to the selected subset of BPMN.
4. Implement the DSL and the model transformation.

5. Validate the approach on a case study.
6. Document the design, implementation, and evaluation in the final report.

4 Deliverables

The following deliverables are due at the end of the project:

Final report Please have a look at the submission requirements in the following section. The final report should consist of an introduction, a (theoretical) background section, sections describing the work packages and their outcomes, and a conclusion. The report must be written in English.

Source code Complete, documented source code with a reproducible evaluation.

Presentation At the end of the thesis, a 20-minute presentation must be given in a thesis defense seminar. It should give an overview and discuss the most important highlights of the work.

References

- [1] David A. Basin, Søren Debois, and Thomas T. Hildebrandt. On purpose and by necessity: Compliance under the GDPR. In Sarah Meiklejohn and Kazuo Sako, editors, *Financial Cryptography and Data Security - 22nd International Conference, FC 2018, Nieuwpoort, Curaçao, February 26 - March 2, 2018, Revised Selected Papers*, volume 10957 of *Lecture Notes in Computer Science*, pages 20–37. Springer, 2018.
- [2] Object Management Group. Business process model and notation (BPMN), version 2.0, 2011. <https://www.omg.org/spec/BPMN/2.0/>.

Thesis Submission Guidelines

Submission Requirements

- **Soft copy:** Submit via email to cse@vgu.edu.vn.
- **Hard copies:** Submit two (02) printed copies to the CSE Office (Room 511, Cluster 3).
- The Thesis cover requires at least 3 days to complete. Please make sure to prepare it at the photocopy shop as soon as possible.

Submission Timeline

- You may submit your Final Thesis Report (soft + hard) up to one week before the deadline.
- Early or late submissions will **NOT** be accepted.

Thesis Title Changes

- If you wish to change your Thesis title, please submit a “Request for Changing Thesis Title” by yourself.
- The request must be sent to us within the first four weeks of the preparation period.

Additional Resources

- Required forms and the Thesis template can be found at [Thesis Resources Link](#).
- Please read the Thesis Guidelines carefully.

Intellectual Property

Project: A Domain-Specific Language for Privacy-Aware Business Processes

Assessor: _____

Student: _____

The student participates in this project at Vietnamese-German University (VGU) as a collaborator under the supervision of the assessor in the course of their studies towards the degree named above. The results produced during the collaboration, including the deliverables listed in Section 4, in particular the source code and the final report, constitute the “Research Results” and are governed by the following terms:

Confidentiality The student keeps the Research Results, together with any other information from the supervising research group at VGU that is declared confidential, confidential. This obligation persists until the student is informed that the intellectual property rights have been protected (e.g. through patent applications or other adequate measures) or that no protection is sought, and in no case for longer than twelve (12) months after completion of the collaboration.

Assignment of rights The student assigns to VGU their rights to the Research Results, including inventions and works protected by copyright. In particular, all rights to the commercial exploitation of the Research Results are ceded to VGU.

Patent cooperation If a patent application is filed for an invention based on the Research Results, the student provides all necessary signatures and remains available to support the application process, e.g. to respond to questions from patent examiners.

By signing below, the parties agree to the terms set out above.

Vietnamese-German University (VGU)

Place, Date

The assessor

The Student

Place, Date

The student